



A Multi-center Dental Panoramic Radiography Image Dataset for Impacted Teeth, Periodontitis, and Dental Caries: Benchmarking Segmentation and Classification Tasks

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Abstract

Panoramic radiography imaging plays a crucial role in the diagnostic process of dental diseases. However, current artificial intelligence research datasets for panoramic radiography dental image processing are often limited to single-center and single-task scenarios, making it difficult to generalize their results. To address this, we present a multi-center, multi-task labeled dataset. In this study, our dataset comprises three datasets obtained from different hospitals. The first set has 4940 panoramic radiography images and corresponding labels from the Stomatological Hospital of the General Hospital of Ningxia Medical University. The second set includes 716 panoramic radiography images and labels from the People's Hospital of Yinchuan City, Ningxia. The third dataset contains 880 panoramic radiography images and labels from a hospital in Shenzhen, Guangdong Province. This comprehensive dataset encompasses three types of dental diseases: impacted teeth, periodontitis, and dental caries. Specifically, it comprises 2555 images related to impacted teeth, 2735 images related to periodontitis, and 1246 images related to dental caries. In order to evaluate the performance of the dataset, we conducted benchmark tests for segmentation and classification tasks on our dataset. The results show that the presented dataset could be effectively used for benchmarking segmentation and classification tasks critical to the diagnosis of dental diseases. To request our multi-center dataset, please visit the address: <https://github.com/qinxin99/qinxini>.

Keywords Dataset · Dental image · Panoramic radiography image · Impacted teeth · Periodontitis · Dental caries

Introduction

With the growing emphasis on dental health, the technology for diagnosing and treating dental diseases has developed rapidly over the past few decades, thanks to the utilization of

diverse imaging and diagnostic tools. Accurate recognition of tooth status and segmentation of teeth form the fundamental basis for the diagnosis and treatment of numerous oral diseases [1]. Nonetheless, due to the complex morphology and structure of teeth, the manual interpretation and analysis

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of X-rays necessitate intensive time and effort [2]. To alleviate the laborious manual process, numerous researchers have devised automated tooth segmentation methods to assist doctors in automatically delineating tooth contours.

In the field of dentistry, computers are used to assist doctors in identifying teeth in panoramic radiography images. This technology not only segments teeth but can also detect specific teeth, such as dental implants. Recent works in the field include Otayk's use of convolutional networks for tooth detection [3]. Muramatsu's classification of tooth types and conditions from panoramic radiography images [4], and Jader's utilization of unsupervised learning methods to segment each tooth in panoramic radiography datasets that are publicly available [5]. Tuzoff [6] also used convolutional networks for tooth detection and numbering, while Fukuda focused on detecting vertical root fracture (VRF) on panoramic radiography [7]. One of the larger dental databases is the Tufts Dental Database [8], which includes panoramic radiographs, segmented masks outlining abnormalities, gray-scale eye-tracking gaze plots, color-quantized eye-tracking gaze plots, teeth masks, and the maxillomandibular region of interest. Rubiu et al. [9] utilized the Tufts Dental Database to design a segmentation model for teeth identification in panoramic dental radiography based on Mask-RCNN. The objective of us is to provide a new multi-center dataset for subsequent segmentation and classification research.

However, AI technology for dentistry confronts several challenges. Many methods suffer from limited generalization ability due to the small size of the existing datasets and the lack of data diversity, thereby limiting the applicability of the segmentation methods to healthy cases or a small number of problematic cases. Additionally, the majority of existing datasets are collected from a single center, resulting in the overfitting of center-specific characteristics during model training and thus hindering generalization to data from different centers [3]–[8]. To solve this problem, we provide a multi-center, multi-task labeled dataset. Table 1 provides a comparison between our dataset and other similar datasets.

Materials and Methods

Materials

Image Acquisition

This study consists of three datasets. Dataset 1 consists of retrospectively collected oral panoramic images from adult patients treated at the stomatological hospital of the General Hospital of Ningxia Medical University. All images in this dataset were captured using the same digital panoramic X-ray machine (Instrumentarium Dental OP200D) and have a resolution of 2976×1536 pixels. The panoramic film included in Dataset 2 was sourced from the People's Hospital of Yinchuan City, Ningxia. The X-ray machines used for capturing these images were Vera-viewepocs and ORTHOPHOS XG, with pixel dimensions of 2893×1480 and 2440×1280 . Dataset 3 was collected from a hospital in Shenzhen, Guangdong Province. All images in this dataset were acquired using a Sirona X-ray machine, possessing a resolution of 2440×1292 pixels.

Impacted teeth encompass embedded and impacted permanent teeth, as well as supernumerary teeth. The scope of caries includes but not limited to primary caries and other forms of caries. An image is considered indicative of periodontitis if it displays at least three consecutive permanent teeth in a single arch or exhibits periodontal lesions [10]. Moreover, in addition to affecting the incisors and first molars, the image must show the loss of adjacent attachment in at least two permanent teeth. Images of patients below the age of 18, images with excessive noise or blurred vision, and images with severe distortion are excluded from the study.

The acquisition process of this dataset is fully ethical and has been approved by the Medical Ethics Review Committee of the General Hospital of Ningxia Medical University, with the number KYLL-2021066.

Table 1 Comparison with other similar datasets

Authors	Variable detected	Target task	Total images	Number of centers	Labeled teeth	Teeth classification labels	Labeled abnormalities
Oktay [3]	Teeth	Detection	100	1	✓		
Muramtsun [4]	Teeth	Classification	100	1	✓		
Jader [5]	Teeth	Segmentation	1500	1	✓		
Tuzoff [6]	Teeth	Detection	1352	1	✓		
Fukuda [7]	Endodontic	Detection	300	1			✓
TDD [8]	Teeth, Abnormalities	Segmentation	1000	1	✓		✓
Our	Teeth, Abnormalities	Classification, Segmentation	6580	3	✓	✓	✓

Image Annotation

Medical image annotation serves as the fundamental component of almost all deep learning methods. In this experiment, all images were converted to NIFTI format using DICOM Converter software. The unified labeling software, ITK-SNAP (Fig. 1), was utilized for manual annotation. Two independent dentists, one of them is a university professor with 25 years of clinical experience, and the other is an associate professor at the university with 18 years of work experience, performed the manual annotation by outlining closed curves along the contours of impacted teeth, areas affected by caries, and positions of bone absorption in the upper and lower alveolar bone in cases of periodontitis. These annotations were recorded in the appropriate format.

To minimize diagnostic errors, two additional dentists conducted a review and made necessary modifications to the annotated images, including additions, deletions, and confirmations. They are associate professors at the university, both with 15 years of clinical experience, and did not exclude any “lesions” or images based on differing diagnostic opinions. We recorded and used the final annotation results in the model. It should be noted that due to the retrospective nature of this experiment, some images may lack complete clinical diagnosis or supporting results from instrumental examinations. Therefore, the final manual annotations solely serve as essential references for training and researching the model and may not necessarily represent definitive facts. The diagnostic model employs a collection of common points that identify all or most of the labels as the diagnostic criteria [11]. Figure 2 illustrates the specific annotation methodology and the resulting labeled image.

From Table 2, we can see, among the images, a total of 2555 panoramic films were diagnosed with impacted teeth, encompassing supernumerary teeth in the upper anterior

teeth area, impacted third molars, and other embedded impacted teeth. A total of 2735 panoramic films were diagnosed with periodontitis. We note that the severity of periodontitis was not differentiated in this experiment. Regarding caries, a total of 1246 panoramic films were identified to contain carious lesions. No distinction was made based on the location of the lesion or the degree of disease progression.

Image Grouping

To minimize the overfitting problem when the training dataset is relatively small, the clinical practicality of the deep learning segmentation model was assessed. The datasets of the three centers are divided according to Table 3, test2 and the training set are from the same center, and the remaining three are cross-center testing sets from different centers.

Datasets 1, 2, 3, and 4 were combined and randomly divided into two mutually exclusive sets for classification tasks—the training group (consisting of 4604 images) and the testing group (consisting of 1976 images) with a 7:3 ratio according to Table 4. The training group was used for machine learning purpose, including tuning model parameters, and determining the final diagnostic model. The testing group was used to evaluate the performance of the constructed segmentation model.

Methods

Experimental Environment

In our project, we utilized some widely used segmentation and classification algorithms that are commonly employed in benchmarks for segmenting and classifying diverse datasets.

Fig. 1 ITK-SNAP interface diagram

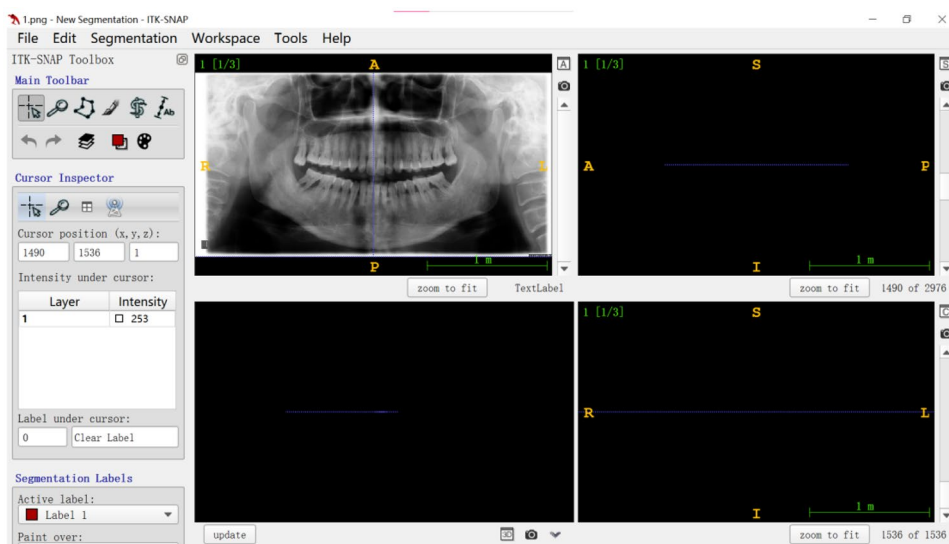


Fig. 2 ITK-SNAP label generation schematic: **a** periodontal disease label schematic, **b** impacted tooth label schematic, **c** caries label schematic

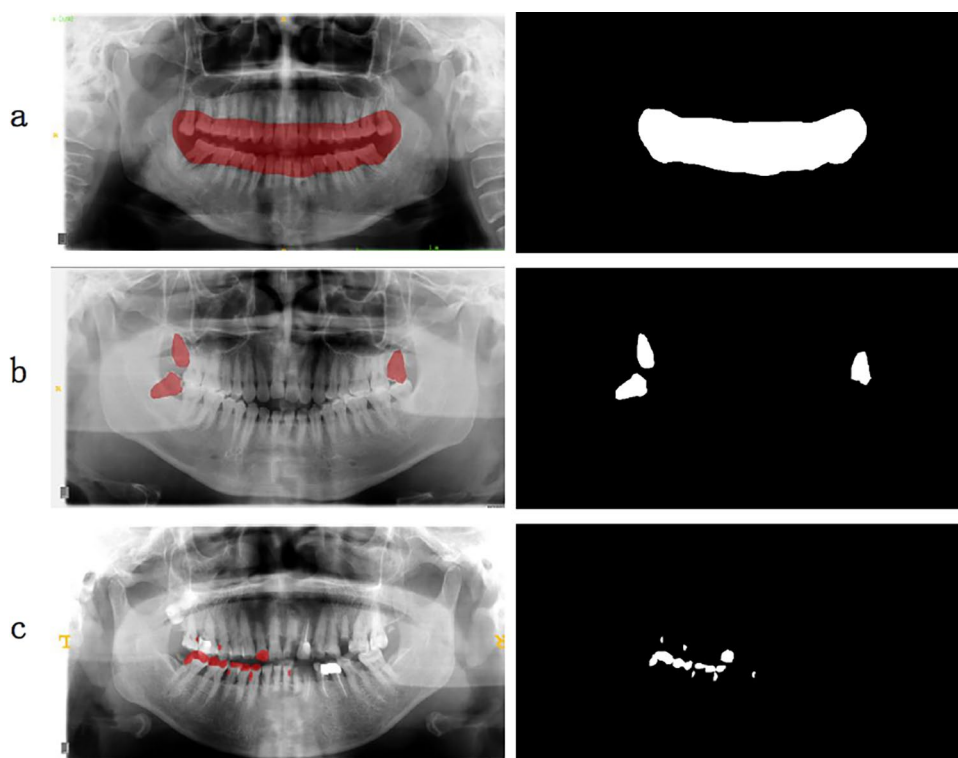


Table 2 Specific classification of dataset images

Category	Data 1	Data 2	Data 3	Data4	Total
Impacted teeth	2210	280	36	41	2567
Periodontitis	2297	320	50	69	2736
Dental caries	1050	134	40	53	1277
Total	5557	734	126	163	6580

Table 4 Data statistics for training and testing groups of classification tasks

Category	Train group	Test group	Total
Impacted teeth	1797	770	2567
Periodontitis	1915	821	2736
Dental caries	894	383	1277
Total	4606	1976	6580

Table 3 Data statistics for training and testing groups of segmentation tasks

Category	Train group	Test1 group	Test2 group	Test3 group	Test4 group	Total
Impacted teeth	2000	280	210	36	41	2567
Periodontitis	2000	320	297	50	69	2736
Dental caries	940	134	110	40	53	1277
Total	4940	734	617	126	163	6580

We hope to use these algorithms to measure the performance of our datasets and provide reference points for future work.

We conducted all segmentation experiments on a NVIDIA Geforce RTX 3090 GPU with 24 GB memory. The software environment is Ubuntu 18.04.1 operating system, Python 3.7.6, Pytorch 1.10.0 deep learning framework, and CUDA 11.3. The segmentation experiment parameter settings are detailed in Table 5.

For each network of classification, we used the Adam Optimizer and trained the model for 20 epochs with a batch size of 128 using the PyTorch framework, with a learning rate of 0.001. The experiment was performed on a server with NVIDIA 3070, using Ubuntu 18.04 operating system.

Segmentation

We evaluated the performance of different deep learning-based algorithms for segmenting teeth from panoramic radiographs. We compared the performance of UNet [12], TransUNet [13], TA-Net [14], CE-Net [15], and

Table 5 Details of the experimental parameter settings

Model	Disease	Epoch	Batch Size	Initial learning rate
Unet [12]	Periodontitis	35	8	1×10^{-5}
	Impacted teeth	50	2	1×10^{-5}
TransUNet [13]	Periodontitis	30	10	1×10^{-2}
	Impacted teeth	30	8	1×10^{-1}
TA-Net [14]	Periodontitis	100	16	2×10^{-4}
	Impacted teeth	100	16	2×10^{-4}
CE-Net [15]	Periodontitis	500	12	2×10^{-4}
	Impacted teeth	500	12	2×10^{-4}
DS-TransUNet [16]	Periodontitis	100	1	1×10^{-3}
	Impacted teeth	100	1	1×10^{-3}

DS-TransUNet [16]. The periodontitis and impacted teeth in the datasets were segmented according to Table 3.

Figure 3 shows the segmentation results of periodontitis disease of the different test sets. The visualization results of these networks are similar to the mask. Figure 4 shows the comparison of the segmentation results of impacted teeth of the different test sets. As can be seen in Fig. 4, CE-Net performs the best, but the tooth in the lower right corner of Fig. 3c fails to be segmented.

Classification

In this study, we conducted an experiment on classification to aid in the diagnosis of various dental diseases. To evaluate the performance of the classification models, we used AlexNet [17], VGG [18], GoogleNet [19], and ResNet [20]. These models have been widely used in image classification tasks and have demonstrated superior performance in various benchmarks. In addition, we have applied the Convolutional block attention module [21] (CBAM) to ResNet.

Fig. 3 Comparison of the segmentation results of periodontitis disease of the different test sets. **a** Example image from Test1, **b** example image from Test2, **c** Example image from Test3, **d** Example image from Test4

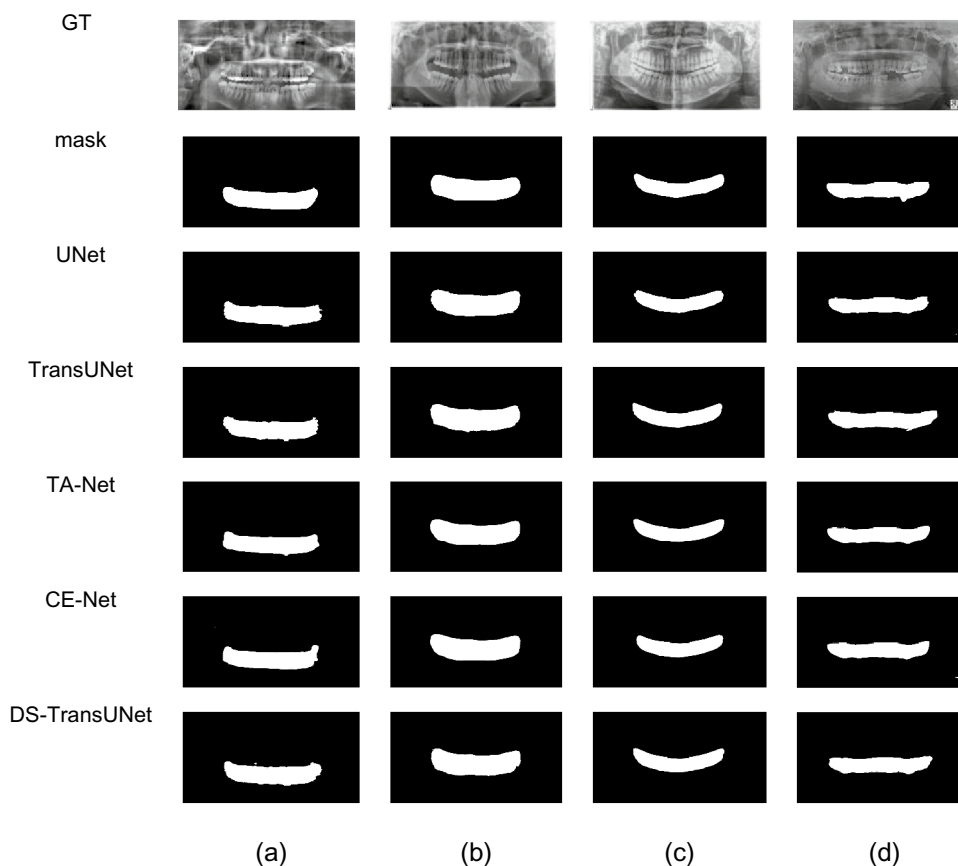
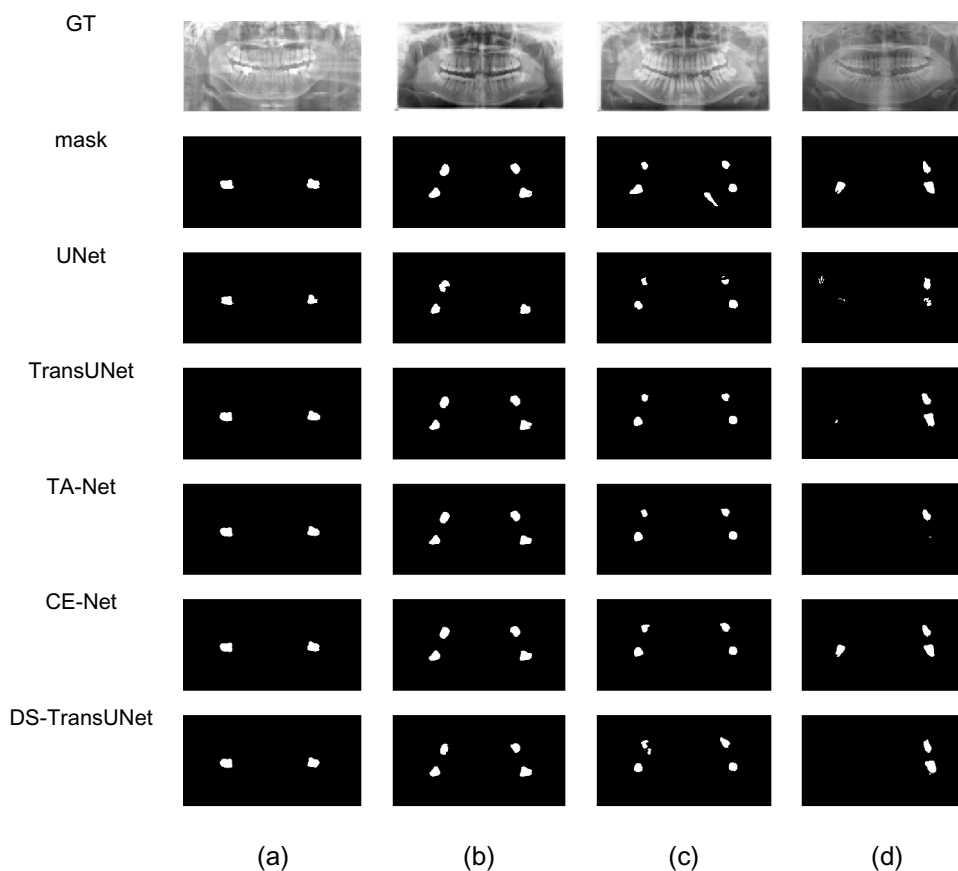


Fig. 4 Comparison of the segmentation results of the impacted teeth of the different test sets. **a** example image from Test1, **b** example image from Test2, **c** example image from Test3, **d** example image from Test4



Before the training process, the images were resized to (224,224) and normalized to ensure consistency and reduce the effects of variations in image size and intensity. During the training process, the loss and accuracy of each epoch were recorded and visualized using Pyplot from Matplotlib (Fig. 5). The results are presented below.

Results

Evaluation Indicators

This paper uses Dice, Intersection over union (IoU), specificity (Spe), sensitivity (Sen), and Hausdorff distance (HD95) to evaluate the performance of networks of segmentation and comparison with different networks. To evaluate the performance of the classification algorithm, we use accuracy, micro_f1, and macro_f1 as performance metrics. Additionally, we performed verification of the datasets from different data centers.

Comparison of Segmentation Networks

Some commonly used metrics are used to evaluate the performance of each segmentation network. As is seen in

Table 6, TA-Net performs the best in the segmentation of periodontitis disease, values in bold indicate that the network performed best on the evaluation indicators. The Dice score, intersection over union (IoU), specificity (Spe), sensitivity (Sen), and Hausdorff distance (HD95) of the model for test 3 are 97.08%, 94.33%, 99.66%, 97.36%, and 0.2078, respectively. In addition, DS-TransUNet has the highest specificity than the other networks.

In our dataset, test2 and the training set come from the same center. To analyze the experimental data statistically, we conducted a two-sided *T* test and used test2 as the internal test set. The other three sets were used as the cross-center test set.

During the periodontitis segmentation experiment, different models were tested and compared across multiple test sets. The UNet model showed no significant differences in any of the indexes tested in test4. However, in test1, there were significant differences in the Dice, IoU, and Sen indexes, whereas in test2, there were significant differences in the Dice, IoU, and Spe indexes. In both cases, the *P*-values were around 0.01. The TransUNet model showed significant differences in all three cross-test sets. In test3, there were no significant differences in Dice, IoU, and HD95 indexes, and the *P*-values for these were 0.83, 0.84, and 0.97, respectively. For the TA-Net model, there were significant differences in all indexes in test1. In test3, there



Fig. 5 Loss and accuracy of each network

Table 6 The performance of algorithms in segmenting periodontitis disease from panoramic radiographs

Evaluation indicators	Test data	UNet [12]	TransUNet [13]	TA-Net [14]	CE-Net [15]	DS-TransUNet [16]
Dice (%)↑	Test1	95.85	94.05	96.88	96.49	92.84
	Test2	96.20	95.56	97.06	96.77	96.31
	Test3	95.06	95.51	97.08	94.48	95.86
	Test4	95.98	94.39	96.90	96.60	95.80
IoU (%)↑	Test1	92.07	88.87	93.96	93.25	87.10
	Test2	92.75	91.51	94.31	93.75	92.93
	Test3	90.74	91.44	94.33	91.19	92.09
	Test4	92.31	89.55	94.00	93.43	91.97
Spe (%)↑	Test1	99.54	99.21	99.64	99.59	99.63
	Test2	99.48	99.38	99.60	99.57	99.77
	Test3	99.22	99.55	99.66	99.31	99.80
	Test4	99.45	99.49	99.53	99.56	99.71
Sen (%)	Test1	96.24	96.12	97.31	97.03	89.94
	Test2	97.13	96.98	97.83	97.53	94.84
	Test3	97.67	95.32	97.36	97.44	93.67
	Test4	97.33	93.87	98.40	97.43	94.49
HD95 (mm)↓	Test1	0.7278	0.8441	0.2972	0.3697	1.6160
	Test2	0.6343	0.4011	0.2025	0.2574	0.1321
	Test3	0.6273	0.3980	0.2078	2.1903	0.1434
	Test4	1.1426	1.7350	0.2333	1.5609	0.2843

were significant differences in the Spe index, and in test4, there were significant differences in the Spe and Sen indexes, with P -values of 0.04, 0.01, and 5.15×10^{-5} , respectively. The CE-Net model showed no significant differences in the indexes tested in test3. However, in test1, there were significant differences, and in test4, the P -value for the HD95 index was 6.19×10^{-44} which was significantly different from the test2. Finally, the DS-TransUNet model showed differences in all test sets, except for the Spe index in test3, where there was no significant difference. The Spe, Sen, and HD95 indexes of test4 had P -values of 0.76, 0.09, and 0.33, respectively, whereas the P -value of the Dice index in test3 was 0.24. From these results, we can conclude that TA-Net and CE-Net performed the best.

We also list the metrics across all models we trained to segment the impacted teeth in Table 7. For each model, each test data is compared horizontally with different evaluation metrics, and the best performance is indicated by bold values. The CE-Net is very similar to the TA-Net in comparison to all indices. For test 2, in Dice and IoU CE-Net is 0.74% and 1.42% higher than TA-Net, while in Sen, TA-Net is 3.18% higher than CE-Net. In DS-TransUNet, Spe is very close to CE-Net and HD95 is 3.6815, the lowest value. Considering the segmentation visualization results and key segmentation metrics, we conclude that TA-Net and CE-Net perform better in the segmentation of impacted teeth.

During the impacted tooth segmentation experiment, the UNet model indexes differed in all test sets except for HD95 of test3 and Spe and HD95 of test4. The values for these were 0.08, 0.51, and 0.56, respectively. The TransUNet model showed significant differences in test1 across the test set, while the Sen model had significant differences in test4, with a P -value of 0.02. In test3, there were significant differences between Sen and HD95, with P -values of 0.11 and 0.13. The TA-Net model showed significant differences in Dice and IoU indexes in each test set, with P -values of test1 and test3 at around 0.002 and test4 at around 0.02. In test4, there were no significant differences in the indexes of the CE-Net model. However, there were significant differences in Dice, IoU, and HD95 in test1, with P -values of 0.020, 0.004, and 0.009, respectively. There were also significant differences in Dice, IoU, and Spe in test3, with P -values of 0.002, 0.001, and 0.001, respectively. The DS-TransUNet model indexes showed significant differences in each test set, except for the Spe of test4, where the value of P was 0.11. The test sets of each center showed obvious differences, indicating that the model needs further improvement.

Comparison of Classification Networks

The results about the comparison of classification networks are given in Table 8. The mixed dataset contains all four test sets mixed together. The results show that the performance

Table 7 The performance of algorithms in segmenting impacted teeth from panoramic radiographs

Evaluation indicators	Test data	UNet [12]	TransUNet [13]	TA-Net [14]	CE-Net [15]	DS-TransUNet [16]
Dice (%)↑	Test1	57.94	68.03	81.99	84.00	70.30
	Test2	74.56	79.50	86.70	87.44	85.85
	Test3	35.70	66.06	71.51	68.76	68.77
	Test4	63.52	70.48	76.79	81.45	73.54
IoU (%)↑	Test1	45.57	54.48	72.52	75.19	58.29
	Test2	63.48	67.64	78.44	79.86	76.64
	Test3	27.05	54.21	60.75	59.93	56.74
	Test4	52.67	60.68	67.95	74.10	62.96
Spe (%)↑	Test1	99.51	99.58	99.74	99.79	99.79
	Test2	99.64	99.66	99.78	99.81	99.93
	Test3	99.31	99.56	99.63	99.67	99.79
	Test4	99.61	99.68	99.71	99.79	99.87
Sen (%)↑	Test1	78.08	91.02	94.75	92.45	68.03
	Test2	87.27	94.60	96.90	93.72	80.96
	Test3	62.38	87.25	90.23	83.15	67.27
	Test4	76.25	81.51	90.05	90.94	68.76
HD95 (mm) ↓	Test1	134.6485	52.4072	38.5784	41.7551	18.6033
	Test2	53.2025	20.9426	19.9986	20.7210	3.6815
	Test3	166.7866	41.1187	53.1968	83.1911	17.0235
	Test4	88.4176	62.8554	63.9953	47.8703	16.0875

Table 8 Performance metrics on different datasets

Evaluation indicators	Test data	Alexnet [17]	VGG [18]	Googlenet [19]	Resnet [20]	Resnet with CBAM [21]
Accuracy (%)↑	Test1	77.83	63.35	81.45	97.74	87.55
	Test2	78.49	83.87	91.40	98.92	94.09
	Test3	78.95	65.79	84.21	97.37	92.11
	Test4	72.00	82.00	86.00	98.16	88.00
	Mixed	74.39	72.42	85.37	98.33	87.55
Micro_F1 (%)↑	Test1	81.67	73.33	82.61	97.77	89.14
	Test2	86.11	88.89	92.50	99.46	95.24
	Test3	79.17	93.75	85.42	97.50	91.67
	Test4	79.17	81.25	93.75	98.21	93.75
	Mixed	74.23	75.54	85.35	98.56	89.14
Macro_F1 (%)↑	Test1	80.38	72.25	82.49	97.01	88.48
	Test2	85.57	88.91	90.47	99.31	95.03
	Test3	77.67	92.93	84.72	96.83	91.11
	Test4	74.81	79.19	91.11	97.57	93.06
	Mixed	69.89	75.89	82.50	98.23	88.48

of ResNet is relatively better. To other algorithms, the accuracy is less than ResNet; according to the accuracy curve, we can notice that there may exist the problem of overfitting. A

two-sided *T* test was conducted on the data in Table 8, using each test set and the mixed test set. There was no significant difference at the *P*-value of 0.05.

Discussion

Based on the benchmark results obtained, we observed that there is a potential inconsistency in the accuracy achieved across different test sets. This inconsistency may be due to the inherent variability in the domains of the datasets obtained from distinct centers. The panoramic radiography images obtained from diverse domains may possess disparities in image quality, resolution, capturing devices, and capturing conditions, which can noticeably impact the effectiveness of the training process and ultimately the results obtained. Hence, conducting research on domain adaptation becomes crucial to address this issue, especially in the context of our multi-center dataset.

Compared to other single-center datasets, our dataset is multi-center and contains annotations for multiple tasks. Oktay [3], Muramtsum [4], Järder [5], and Tuzoff [6] annotated their datasets for Teeth, while Fukuda [7] annotated their dataset for abnormalities. Our dataset and TDD [8] dataset both annotated for Teeth and abnormalities. However, the TDD [8] dataset has an advantage over ours as it includes eye-tracking in the dataset, which our dataset lacks. In this regard, TDD [8] performs better. Our advantage lies in the fact that we have a larger dataset. TDD [8] dataset has 1000 teeth data, while our work has 6580. Additionally, the TDD [8] dataset was collected from a single center, while our dataset was collected from three different centers. This gives our dataset an advantage in multi-center domain generalization research.

The limitation of this study lies in the current segmentation algorithm's inability to accurately delineate caries lesions of this dataset. Should future endeavors aim to segment these lesions, substantial model enhancements will be necessary. Another limitation is that there is no clear statistics on the total number of diseased teeth in the image, which is the focus of future research.

Conclusion

In this study, we present a multicenter panoramic radiography dental image dataset that includes annotations for three different dental diseases. We have conducted a series of baseline experiments for two tasks, including the segmentation of various tooth lesion shapes, and the classification of different tooth diseases.

For future work, we plan to expand the dataset by including additional disease categories. We also aim to enhance the segmentation performance across different centers and minimize the variations in segmentation results. This dataset can serve as a valuable resource for

investigating domain generalization challenges across different dental centers. Furthermore, we have identified the complexities involved in segmenting caries in panoramic radiography oral images. Our dataset can be used in conjunction with existing datasets to regularize models and improve their performance under various conditions. Our dataset holds the potential for training neural networks for segmentation and classification tasks that enable the accurate disease diagnosis and precise localization of pathological lesions. Consequently, the presented dataset can be used for designing and developing automated oral healthcare systems.

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Data Availability This dataset is made freely available and can be accessed on the GitHub repository - <https://github.com/qinxin99/qinxini>. Please see Table 2 for details to the data.

Declarations

Ethics Approval The acquisition process of this dataset is fully ethical and has been approved by the Medical Ethics Review Committee of the General Hospital of Ningxia Medical University, with the number KYLL-2021066.

Consent to Participate and for Publication Informed consent was obtained from all individual participants who contributed to the dental panoramic radiography image dataset.

Competing Interests The authors declare no competing interests.

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